

Disk Management & Partitioning

Project Overview:

This project demonstrates how to manage disks, create virtual hard drives, partition, format, and optimize storage in **Windows 10 Pro**.

Disk management is an essential IT Support skill used for troubleshooting low disk space, configuring new drives, and maintaining system performance.

Objectives

- Create and attach a **Virtual Hard Disk (VHD)**.
- Partition and format the new drive.
- Perform volume operations (extend and shrink).
- Run **Disk Cleanup** to free storage.
- Optimize drives for better performance.

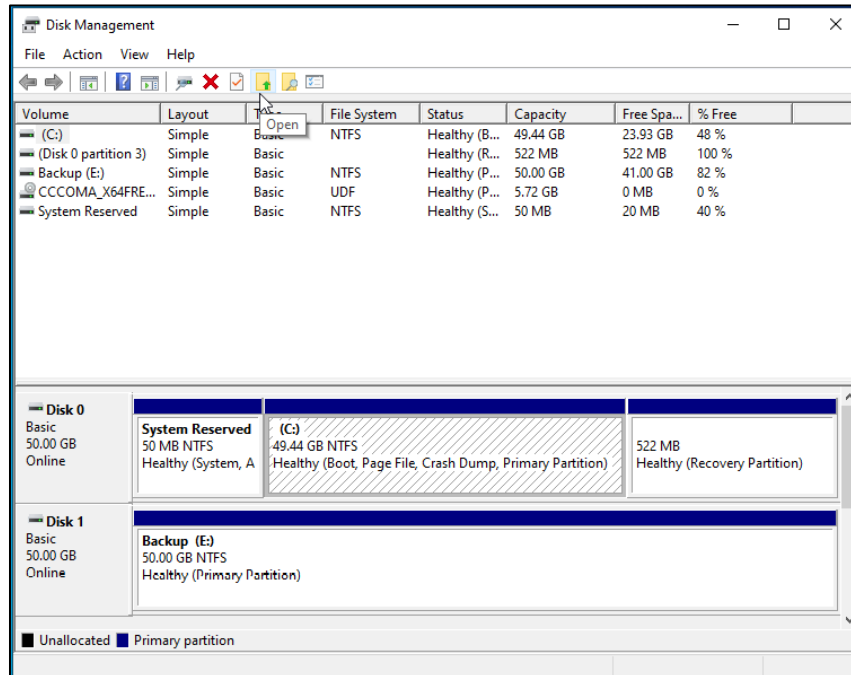
Tools & Skills Used

1. **Windows 10 Pro** Virtual Machine.
2. Disk Management Console (diskmgmt.msc).
3. **Disk Cleanup & Optimize Drives** utilities
4. **NTFS File System & Partitioning** concepts
5. Skills: Troubleshooting, Storage Management, Documentation

Steps & Implementation

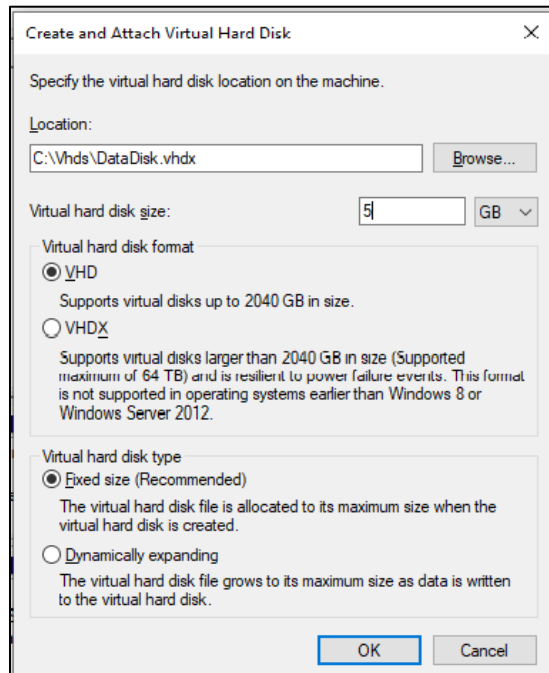
1. Open Disk Management

- Used Win + X → Disk Management.
- Viewed existing drives and partitions.



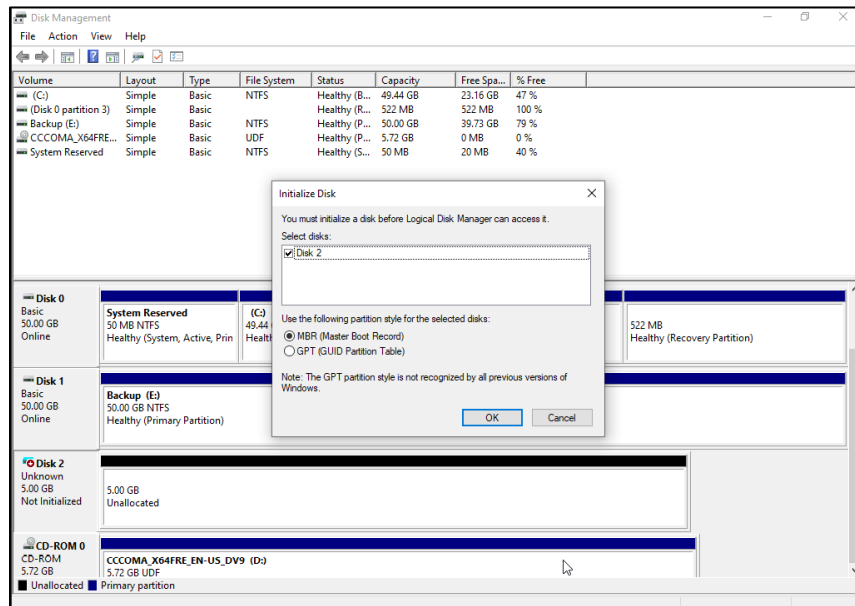
2. Create a Virtual Hard Disk (VHD)

- Created a 5 GB VHDX file at C:\Vhds\DataDisk.vhdx.



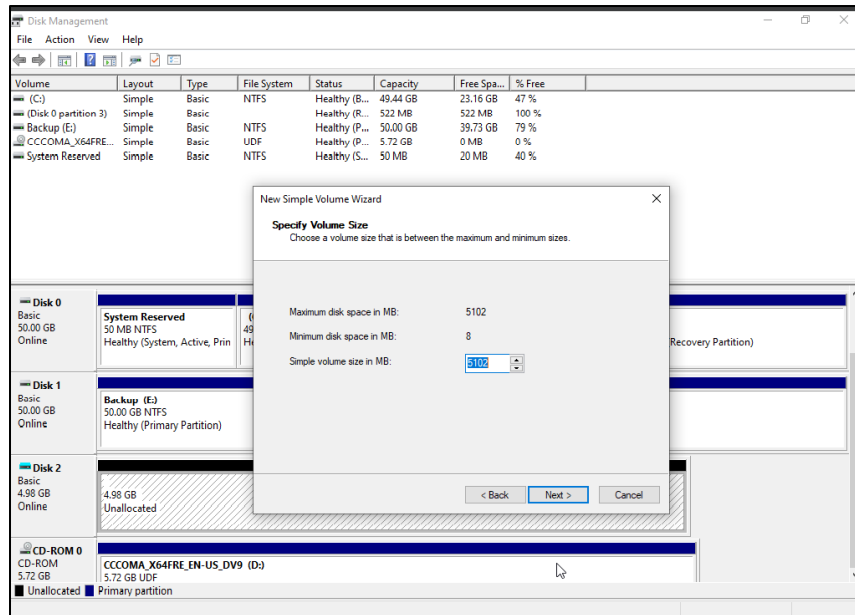
3. Initialize the Disk

- Initialized new disk using GPT partition style.



4. Create and Format Partition

- Created a New Simple Volume.
- Assigned drive letter **F:**.
- Formatted with **NTFS** and named it DataDisk.

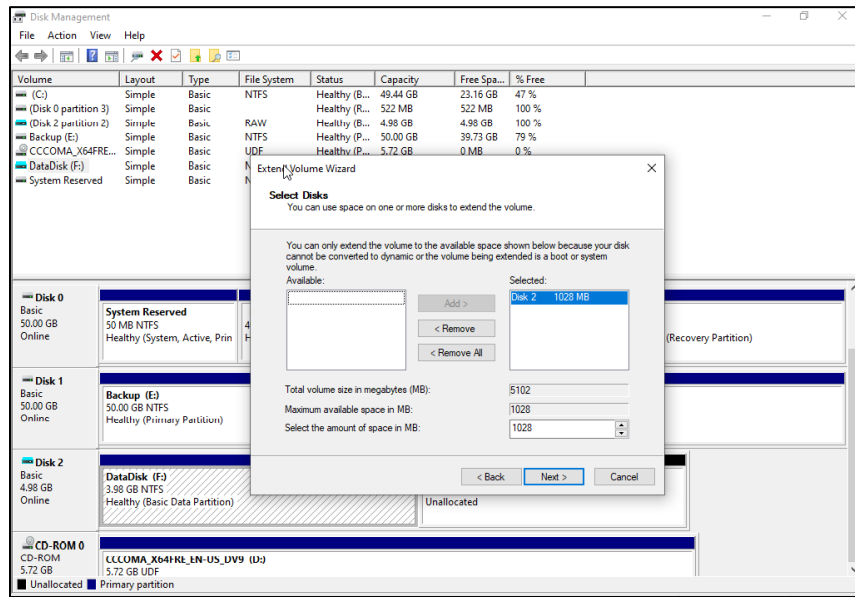


5. Perform Volume Operations

- Shrink Volume → Reduced by 1 GB.

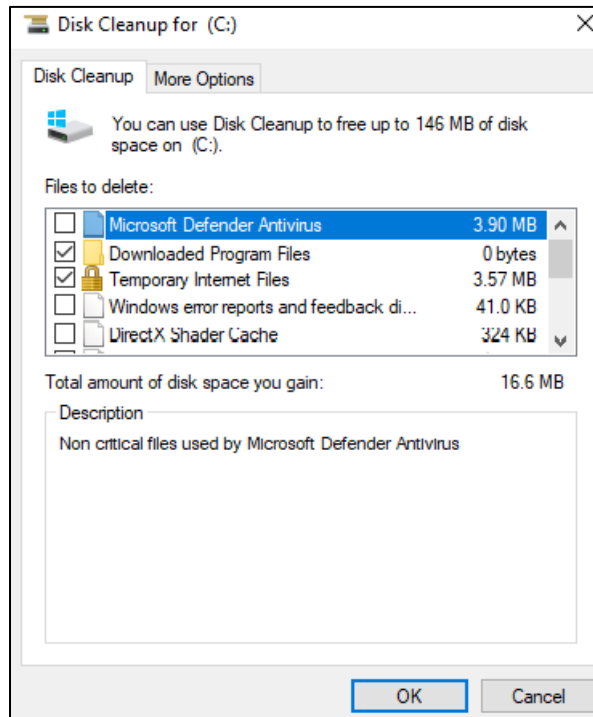
Disk 0	Basic 50.00 GB Online	System Reserved 50 MB NTFS Healthy (System, Active, Prin	(C:) 49.44 GB NTFS Healthy (Boot, Page File, Crash Dump, Primary Partition)	522 MB Healthy (Recovery Partition)
Disk 1	Basic 50.00 GB Online	Backup (E:) 50.00 GB NTFS Healthy (Primary Partition)		
Disk 2	Basic 4.98 GB Online	DataDisk (F:) 3.98 GB NTFS Healthy (Basic Data Partition)	1.01 GB Unallocated	

- Extend Volume → Added free space back.



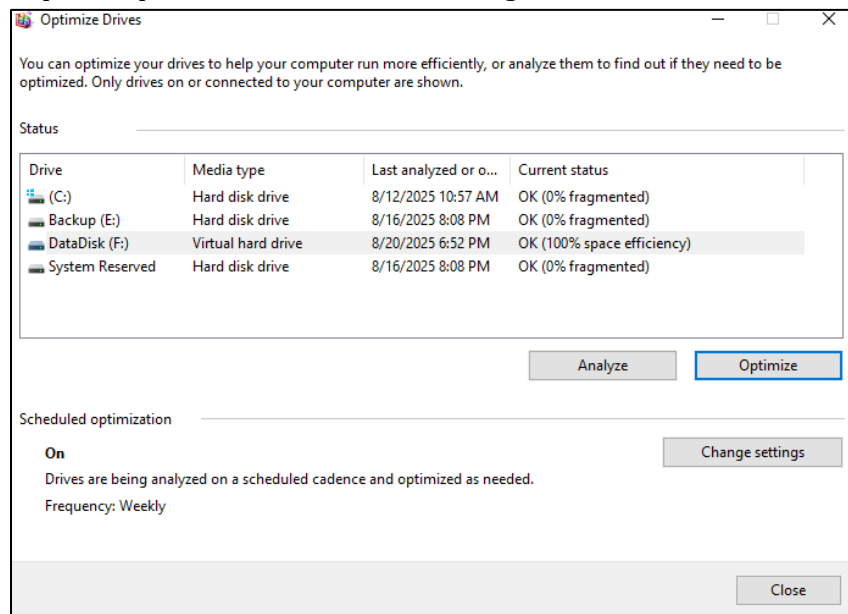
6. Run Disk Cleanup

- Cleared **temporary files** and **Windows Update cache**.



7. Optimize Drives

- Ran **Defragment** and **Optimize** tool on DataDisk.
- Improved performance and reduced fragmentation.



Result:

- ✓ Successfully created and managed a new virtual hard disk.
- ✓ Demonstrated disk partitioning, formatting, extending, and shrinking.
- ✓ Optimized drive performance and freed space using cleanup tools.
- ✓ Showcased **IT Support troubleshooting for low disk space scenarios**.

What I Learned:

- Learned how to simulate adding a new drive using VHDs.
- Practiced **partition management** and NTFS formatting.
- Demonstrated **troubleshooting skills** for common storage issues.